

Sigma algebra : Let X be any set.

A sigma algebra Σ is a subset of $\mathcal{P}(X)$ satisfying

- (i) $\emptyset \in \Sigma$
- (ii) $A \in \Sigma \Rightarrow A^c (= X \setminus A) \in \Sigma$
- (iii) $A_n \in \Sigma, n \in \mathbb{N} \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \Sigma$

Examples : $\Sigma = \{\emptyset, X\}$

$\Sigma = \mathcal{P}(X)$

$\Sigma = \{\emptyset, A, A^c, X\}$, where $A \neq \emptyset, X$.

If $\mathcal{A} \subseteq \mathcal{P}(X)$, then the σ -algebra generated by $\mathcal{A}$ is the smallest σ -alg. containing $\mathcal{A}$.

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Measure

Let X be any set and Σ be a σ -algebra on X . A measure $\mu : \Sigma \rightarrow [0, \infty]$ s.t.

- (i) $\mu(\emptyset) = 0$
- (ii) $A \subseteq B, A, B \in \Sigma \Rightarrow \mu(A) \leq \mu(B)$
- (iii) $\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$, if $\{A_n\}_{n=1}^{\infty}$ is a collection of pairwise disjoint sets in Σ .

Example : $\Sigma = \mathcal{P}(X)$, μ = counting measure.

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Measurable sets

Let μ be an outer measure on a set X .
A subset E of X is said to be a measurable set (w.r.t. μ) if for any $A \subseteq X$,

$$\mu(A) = \mu(A \cap E) + \mu(A \setminus E)$$

- $\Sigma = \{E \subseteq X : E \text{ is measurable}\}$
is a σ -algebra on X (Verify)
- $\mu|_{\Sigma}$ is a measure on (X, Σ) . (Verify)
- $\mu(E) = 0 \Rightarrow E$ is measurable.

Pf: $\mu(E) = 0 \Rightarrow \mu(A \cap E) = 0$
 $\therefore \mu(A \cap E) + \mu(A \cap E^c) = \mu(A \cap E^c) \leq \mu(A)$

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$$\text{Also, } \mu(A) = \mu((A \cap E) \cup (A \cap E^c)) \\ \leq \mu(A \cap E) + \mu(A \cap E^c)$$

$$\therefore \mu(A) = \mu(A \cap E) + \mu(A \cap E^c) \\ \Rightarrow E \text{ is measurable}$$

If we take μ to be the Lebesgue outer measure on $\mathbb{R}$, and let $\mathcal{M}$ to be the σ -algebra of Lebesgue measurable sets, we denote by $\mathcal{L}^1$ the Lebesgue measure on $\mathbb{R}$, which is the restriction of μ on $\mathcal{M}$.

Fact: Open sets are Lebesgue measurable.

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Borel σ -algebra on $\mathbb{R}$, $\mathcal{B}$ is the σ -algebra generated by the open sets in $\mathbb{R}$.

- Since open sets are Lebesgue measurable, $\mathcal{B} \subseteq \mathcal{M}$.
- In fact, $\mathcal{B}$ is a proper subset of $\mathcal{M}$, i.e., $\exists$ a Lebesgue measurable set E which is not Borel measurable ($E \notin \mathcal{B}$).
- There exists a non-measurable set (i.e. $A \subseteq \mathbb{R}$ s.t. $A \notin \mathcal{M}$).

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Hausdorff Measure & Hausdorff Dimension

s-dimensional Hausdorff measure

Let $s \geq 0$ be any real number. For any subset F of $\mathbb{R}^n$, we'll define $\mathcal{H}^s(F)$ to be the s-dimensional Hausdorff measure as follows.

First, for $\delta > 0$, we define

$$\mathcal{H}_\delta^s(F) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam } U_i)^s : \{U_i\}_{i=1}^{\infty} \text{ is a } \delta\text{-cover of } F \right\}$$

($\{U_i\}_{i=1}^{\infty}$ is a δ -cover of F means $F \subseteq \bigcup_{i=1}^{\infty} U_i$ and $\text{diam } U_i \leq \delta$)

$$\text{diam } U = \sup \{d(x, y) : x, y \in U\}$$

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Note that if $0 < \delta_1 < \delta_2$, then $\mathcal{H}_{\delta_2}^s(F) \leq \mathcal{H}_{\delta_1}^s(F)$

$\therefore$ As δ decreases, $\mathcal{H}_{\delta}^s(F)$ increases.

So, we define

$$\mathcal{H}^s(F) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_{\delta}^s(F) = \sup_{\delta > 0} \mathcal{H}_{\delta}^s(F).$$

(The limit may be $+\infty$).

Exercise: Prove that $\mathcal{H}^s$ is an outer measure on $\mathbb{R}^n$.

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